Dissertation Proposal Form

This form must be approved before a student can start working on the dissertation. The student must ensure that all academic and administrative requirements are fulfilled. The Academic Department is responsible for assigning a supervisor/mentor to the student. Once the proposal is approved, this signed form will be sent to and to the student.

SECTION A: STUDENTS DETAILS

Name and Surname: Alban Krasniqi

ID: _____

Email: _____

Program: Bachelor in Computer Science, UNI–Universum International College

SECTION B: PROPOSAL

Type below each topic

Proposed Dissertation Title: The title (tentative) should be precise to allow the reader to comprehend the subject matter of the work.

Asymmetric Self-Play in the Berlin Defence (Vetë-luajtja asimetrike në Mbrojtjen e Berlinit)

Research Question or Hypothesis: State clearly what your research interest is. Define your research question/hypothesis and the main objectives of your research (*100 words*).

The study asks whether an asymmetric search objective, learned during self-play, raises Black's FIDE score in the Berlin Defence. Against the same standard White, does variant Black (two networks, Lc0 DrawScore and parent-relative first-play urgency) score higher than standard Black (one network, Q = W − L and classic first-play urgency), when both train from the C65 position with an equal game budget? The null hypothesis says the variant score is not higher. The alternative says that it is. A secondary hypothesis concerns decisive games, and a labelled question asks whether a win-only objective raises White's score.

Review of Literature/Background: Describe the literature that will be used and reviewed, how your intended research will complement and expand on the current literature (500 words).

Computer chess offers two lines of work for this study. Shannon (1950) showed how a machine can play chess, and alpha-beta pruning made deep search practical (Knuth & Moore, 1975; Campbell, Hoane, & Hsu, 2002). A second line learns from self-play. Tesauro (1995) reached master strength with temporal-difference learning. Sutton and Barto (2018) state the claim used here: a value exists only relative to the reward, so changing what a draw is worth changes the task.

AlphaGo combined human games, self-play and Monte Carlo tree search (Silver et al., 2016). AlphaGo Zero removed the human games (Silver et al., 2017a). AlphaZero carried the algorithm into chess: the network estimates a policy and a value, and the terminal result is −1, 0 or +1 (Silver et al., 2017b). Both colours share that result, and Sadler and Regan (2019) treat it as the source of style. Move selection uses PUCT (Kocsis & Szepesvári, 2006; Browne et al., 2012). Lai (2015) showed that a shallow network with a small search can already play chess, which is the scale used here.

Leela Chess Zero documents the network and the classic search replicated here in Python (Leela Chess Zero, n.d.). From version 0.30.0 it can rescale the win/draw/loss value with contempt at search time, on a network already trained with the ordinary objective (Leela Chess Zero, 2023). Contempt does not train two specialists who value the draw differently from the first game. League methods show that a partner

need not be a copy of the agent (Jaderberg et al., 2017; Berner et al., 2019; Zahavy et al., 2023). They do not show that two small specialists, trained from one opening, will outscore one network at equal visits.

The Berlin Defence is the domain because draws are common. Kramnik played 1.e4 e5 2.Nf3 Nc6 3.Bb5 Nf6 against Kasparov in London in 2000. Tomašev, Paquet, Hassabis and Kramnik (2022) note that engines then valued the Berlin endgame near +1 for White, while current engines treat it as equal. In a 2018 consultative game, AlphaZero met the Berlin and did not change style against a weaker side (Howell, 2018; Lawson, 2018). Following Leela Chess Zero (2019), every game in this study begins from ECO C65, after those six plies. The Berlin Wall (C67) is a reference, not the start position.

Three premises follow, and three gaps remain. Value depends on the reward (Sutton & Barto, 2018). AlphaZero kept one terminal result for both colours (Silver et al., 2017b), while Lc0 contempt changes the search scalar after training (Leela Chess Zero, 2023). The Berlin is draw-prone, with a written human precedent (Tomašev et al., 2022). Missing are an asymmetric objective during chess self-play, both lines trained from the same position, and an equal-budget match against one standard engine, with the decision rule fixed before the scores are seen. The dissertation closes those gaps at one shallow scale only: a 6×64 network, 120 visits and an equal training budget, not full Leela or Stockfish.

Research Methodology: Clearly indicate the methods and techniques which you are planning to use to gather information and investigate the identified problem.

The study is quantitative and experimental. There are no human participants. The independent variable is the search package used during self-play. The control is one network with the classical scalar $Q = W - L$ and classic first-play urgency. The main treatment is two specialist networks, Lc0 DrawScore and parent-relative first-play urgency. DrawScore is $Q = (W - L) + d_s D$, with $d_s = +1$ for White and $d_s = -1$ for Black. The value head still learns the true win/draw/loss outcome. Asymmetry is applied only when that distribution is converted into the PUCT scalar. A third, labelled engine tests a win-only scalar, $Q = W$, as White against the same standard Black. That match is reported separately and is not the test of the main hypothesis.

All lines share one body: a 6×64 squeeze-and-excitation residual network, 112 input planes, an 1858-move policy, a win/draw/loss head and a moves-left head. Training and matches use 120 Monte Carlo visits. The optimiser is AdamW. Every game starts from the same Berlin position (ECO C65), and the six opening plies are forced. Self-

play does not use human games. Evaluation freezes an equal checkpoint of 3,440 games, with N = 100 games, fixed colours, temperature 1 for eight plies after Berlin and then greedy play. Contempt is off. Each game is stored in Portable Game Notation and is rejected if it does not begin with the Berlin moves. The engines are UCI programs written for this thesis. They replicate the documented Leela classic search in Python; they are not a rebuild of the public C++ binary.

The dependent variable is the mean FIDE game score of the tested side, s = (wins + 0.5 × draws) / N. For the main hypothesis, Δs is the difference between variant Black and standard Black against the same standard White. The decision uses the 95% confidence interval of that difference, estimated from the two independent samples. An interval lying entirely above zero supports the alternative. An interval lying entirely below zero rejects it in the direction of harm. Otherwise the result is inconclusive. A secondary comparison uses the rate of decisive games. A search-only ablation, without retraining, separates DrawScore, the second network and the first-play-urgency rule. After the scores are seen, N, the opening and the visit limit will not be changed. Hardware is one NVIDIA GeForce RTX 4060 (8 GB). The result is limited to this opening, this network size and this budget.

Selected Bibliography: Please indicate a representative list of the main bibliographical sources you intend to utilize for your dissertation (no more than 20). Please provide full citations.

Berner, C., Brockman, G., Chan, B., Cheung, V., Dębiak, P., Dennison, C., Farhi, D., Fischer, Q., Hashme, S., Hesse, C., Józefowicz, R., Gray, S., Olsson, C., Pachocki, J., Petrov, M., Pinto, H. P. d. O., Raiman, J., Salimans, T., Schlatter, J., Schneider, J., Sidor, S., Sutskever, I., Tang, J., Wolski, F., & Zaremba, W. (2019). *Dota 2 with large scale deep reinforcement learning*. arXiv. https://arxiv.org/abs/1912.06680

Browne, C. B., Powley, E., Whitehouse, D., Lucas, S. M., Cowling, P. I., Rohlfshagen, P., Tavener, S., Perez, D., Samothrakis, S., & Colton, S. (2012). A survey of Monte Carlo tree search methods. *IEEE Transactions on Computational Intelligence and AI in Games, 4*(1), 1–43. https://doi.org/10.1109/TCIAIG.2012.2186810

Campbell, M., Hoane, A. J., Jr., & Hsu, F.-h. (2002). Deep Blue. *Artificial Intelligence, 134*(1–2), 57–83. https://doi.org/10.1016/S0004-3702(01)00129-1

Howell, D. (2018). Chess column: Consultative game against AlphaZero (Berlin Defence). *The Sunday Times*.

Jaderberg, M., Dalibard, V., Osindero, S., Czarnecki, W. M., Donahue, J., Razavi, A., Vinyals, O., Green, T., Dunning, I., Simonyan, K., Fernando, C., & Kavukcuoglu, K. (2017). *Population based training of neural networks*. arXiv. https://arxiv.org/abs/1711.09846

Knuth, D. E., & Moore, R. W. (1975). An analysis of alpha-beta pruning. *Artificial Intelligence, 6*(4), 293–326. https://doi.org/10.1016/0004-3702(75)90019-3

Kocsis, L., & Szepesvári, C. (2006). Bandit based Monte-Carlo planning. In J. Fürnkranz, T. Scheffer, & M. Spiliopoulou (Eds.), *Machine learning: ECML 2006* (pp. 282–293). Springer. https://doi.org/10.1007/11871842_29

Lai, M. (2015). *Giraffe: Using deep reinforcement learning to play chess* [Master's thesis, Imperial College London]. arXiv. https://arxiv.org/abs/1509.01549

Lawson, D. (2018). Dominic Lawson v AlphaZero. *The Times / The Sunday Times*.

Leela Chess Zero. (n.d.). *Developer documentation: Architecture, search, network topology*. https://lczero.org/dev/

Leela Chess Zero. (2019). *Train not only from startpos* (Issue #787). GitHub. https://github.com/LeelaChessZero/lc0/issues/787

Leela Chess Zero. (2023, July). *The Lc0 v0.30.0 WDL rescale/contempt implementation*. https://lczero.org/blog/2023/07/the-lc0-v0.30.0-wdl-rescale-contempt-implementation/

Sadler, M., & Regan, N. (2019). *Game changer: AlphaZero's groundbreaking chess strategies and the promise of AI*. New in Chess.

Shannon, C. E. (1950). Programming a computer for playing chess. *Philosophical Magazine, 41*(314), 256–275. https://doi.org/10.1080/14786445008521796

Silver, D., Huang, A., Maddison, C. J., Guez, A., Sifre, L., van den Driessche, G., Schrittwieser, J., Antonoglou, I., Panneershelvam, V., Lanctot, M., Dieleman, S., Grewe, D., Nham, J., Kalchbrenner, N., Sutskever, I., Lillicrap, T., Leach, M., Kavukcuoglu, K., Graepel, T., & Hassabis, D. (2016). Mastering the game of Go with deep neural networks and tree search. *Nature, 529*(7587), 484–489. https://doi.org/10.1038/nature16961

Silver, D., Schrittwieser, J., Simonyan, K., Antonoglou, I., Huang, A., Guez, A., Hubert, T., Baker, L., Lai, M., Bolton, A., Chen, Y., Lillicrap, T., Hui, F., Sifre, L., van den Driessche, G., Graepel, T., & Hassabis, D. (2017a). Mastering the game of Go without human knowledge. *Nature, 550*(7676), 354–359. https://doi.org/10.1038/nature24270

Silver, D., Hubert, T., Schrittwieser, J., Antonoglou, I., Lai, M., Guez, A., Lanctot, M., Sifre, L., Kumaran, D., Graepel, T., Lillicrap, T., Simonyan, K., & Hassabis, D. (2017b). *Mastering chess and shogi by self-play with a general reinforcement learning algorithm*. arXiv. https://arxiv.org/abs/1712.01815

Sutton, R. S., & Barto, A. G. (2018). *Reinforcement learning: An introduction* (2nd ed.). MIT Press.

Tesauro, G. (1995). Temporal difference learning and TD-Gammon. *Communications of the ACM, 38*(3), 58–68. https://doi.org/10.1145/203330.203343

Tomašev, N., Paquet, U., Hassabis, D., & Kramnik, V. (2022). Reimagining chess with AlphaZero. *Communications of the ACM, 65*(2), 60–67. https://doi.org/10.1145/3460349

Research and Writing Timeline: Indicate your expected research timetable: starting date, research time, data analysis time, writing time, editing time, date of first draft submission, expected defense time.

Starting date: 9 September 2026.

Literature review: completed together with the research design, and closed before the protocol was frozen for the draft of 14 September 2026.

Research and implementation: 9–22 September 2026 (standard, variant and win-only / aggressive UCI engines, Berlin opening, and training scripts).

Data collection: fixed-colour matches of 100 games at 120 visits, stored by 23 September 2026.

Data analysis: 23–24 September 2026 (game scores, confidence intervals, search ablation, and Portable Game Notation summary).

Writing time: through 25 September 2026. Editing time: 26 September 2026 to 4 October 2026.

Date of first draft submission: 25 September 2026. Expected defence: October 2026.

Ethics: Are there any ethical issues related to your research, and how will these be addressed to ensure adherence to ethical standards in your study?

There are no human subjects, so informed consent, confidentiality of participants and a right of withdrawal do not arise. The networks will not be trained on human games. The Kramnik–Kasparov games of 2000 are cited as historical background, not used as training data. Approval of the diploma topic is requested through this proposal, as the faculty guide provides. Game scores will not be removed after they are seen, and the sample size, the opening and the visit limit will stay fixed. Negative findings will be reported. There is no external funding and no conflict of interest. Citations follow APA 7. The student remains responsible for the recorded positions, the citations and the numbers in the results. If an editing tool is used, it does not replace that responsibility.

SECTION C: APPROVAL

Role	Name	Signature	Date
Mentor/Supervisor	Prof. Ass. Dr. Rrezart Prebeza		
Dissertation Coordinator			
Academic Department			
Plagiarism reviewer			
AI-content reviewer			

Date: 04/10/2026